

Architecture

The live site is a static Next.js export served from a private S3 bucket through CloudFront. The browser calls a public, response-streaming Lambda Function URL whose container runs FastAPI, LangGraph, and five MCP subprocesses. A LangGraph **supervisor StateGraph** routes each turn to one of four scoped `create_react_agent` specialist subgraphs; a hop cap of 3 prevents routing loops. Specialists dispatch over stdin/stdout JSON-RPC or call the in-process Chroma/BM25 knowledge tools. Mock SaaS state lives in SQLite under Lambda /tmp; durable conversation and interrupt checkpoints live in DynamoDB.

AI Agent Application

The supervisor is a pure router — it uses structured output (`Route{next, reasoning}`) to pick exactly one specialist per hop and never produces user-visible text. Each specialist owns a scoped toolbox, so tool choice stays tight (the Training Coach cannot accidentally file an IT ticket).

- **HR Profile Specialist** — updates across HR Platform, Slack, and Salesforce; doubles as the onboarding planner for open-ended “what do I need to do?” questions.
- **Training Coach** — tracks four mandatory modules in order; never marks a module complete until the user explicitly says they finished it.
- **IT Access Specialist** — role-based recommendations → user picks → manager approval → IT ticket; each write is gated.
- **Knowledge Expert** — hybrid-RAG Q&A over seven policy documents with source attribution; read-only.

Specialists are written as collaborators: they ask one focused question when information is missing, propose changes before writing, and narrate progress briefly. Direct imperatives with full values execute immediately — the HITL gate is the final review, not a substitute for conversation.

Amazon Bedrock Nova is the AWS chat provider and Titan Text Embeddings v2 builds the RAG index. Local development keeps the OpenAI/Ollama provider path.

RAG Pipeline (3 production techniques)

1. **Cosine similarity** — ChromaDB collection configured with `hnsw:space=cosine` so relevance is magnitude-independent.
2. **Contextual chunking** — each chunk is prefixed with its document title and section header before em-

bedding, preserving meaning in isolation.

3. **Hybrid retrieval** — BM25 keyword search and vector semantic search merged via Reciprocal Rank Fusion (equal 0.5/0.5 weighting with `EnsembleRetriever`).

Human-in-the-Loop

Eight destructive tools (`every update_*`, `add_to_slack_channels`, `assign_salesforce_permission_set`, `complete_training_module`, `request_manager_approval`, `submit_it_ticket`) are wrapped at load time with a LangGraph `interrupt()` gate. When a specialist calls one, the graph pauses, the API emits an `approval_required` SSE event, and the frontend renders an inline approval card where the user can review arguments, **edit any string field**, then approve or reject. Only changed keys are sent as `edited.args`; the wrapper merges them (`{**kwargs, **edited.args}`) before invoking the underlying tool. State is checkpointed by DynamoDB on AWS or `MemorySaver` locally, so the approval survives across separate HTTP turns; rejections are fed back as a `[SKIPPED]` tool message the specialist handles gracefully.

Evaluation

A golden-dataset harness (`python -m evals.run_evals`) exercises the full supervisor graph across 15 cases — HR, Training, IT, Knowledge, and routing edges. Each case runs on a fresh `thread_id` (so cases cannot leak state) and is graded by four deterministic hard-gate evaluators — `routing`, `tool_trajectory`, `tool_choice`, `response_contains` — plus an LLM-as-judge scoring response quality (1–5) against a per-case rubric. The runner exits non-zero on any hard-gate failure, so it drops straight into CI; runs are traced to LangSmith when tracing is enabled.

Orchestration

Request flow: `POST /api/chat` → FastAPI invokes the orchestrator → supervisor routes to a specialist → specialist issues tool calls (possibly pausing at an HITL gate) → results stream back as SSE (`agent_handoff`, `tool_call`, `tool_result`, `approval_required`, `text_delta`, `done`). Interrupted runs resume via `POST /api/chat/resume` with the approval decision.

MCP communication: Each of the five servers is a subprocess spawned at startup. `langchain-mcp-adapters` serialises tool calls to JSON-RPC over stdin/stdout; the HITL wrapper sits between the adapter and the specialist so gating is transparent to both sides.

Memory: DynamoDBSaver on AWS (or MemorySaver locally) keeps per-employee conversation history *and* graph interrupt state keyed by `thread_id`, so multi-turn context and paused approvals survive between HTTP requests and, on AWS, Lambda recycling.

AWS Deployment

Terraform provisions CloudFront, a private S3 frontend origin, a public Lambda Function URL, ECR, DynamoDB checkpoints, CloudWatch logs, and an EventBridge warmer. GitHub Actions assumes an IAM role through OIDC only from the public repository's `main` branch and allowed environments. Deployments tag the backend image with the commit SHA, apply Terraform, bake the emitted API URL into the frontend, synchronize S3, invalidate CloudFront, and smoke-test health plus all 20 tools. Live site: d2j7378c90nw96.cloudfront.net.

Trade-offs & Considerations

Assumptions & Trade-offs

- **In-process RAG** — ChromaDB's embedded SQLite is unsafe for cross-process access. Knowledge tools run in the main process, sacrificing isolation to avoid file-lock deadlocks.
- **Ephemeral mock data** — Lambda stores SQLite, Chroma, and BM25 under `/tmp`. New or concurrent execution environments can reset or diverge; only DynamoDB graph checkpoints are durable.
- **Mock MCP servers** — simulate Workday, ServiceNow, Slack, Salesforce, and an LMS. Production

integration needs OAuth, rate limiting, and idempotency guards.

- **Public demo** — the Function URL has no AWS authentication. CORS restricts browsers but does not authenticate callers.

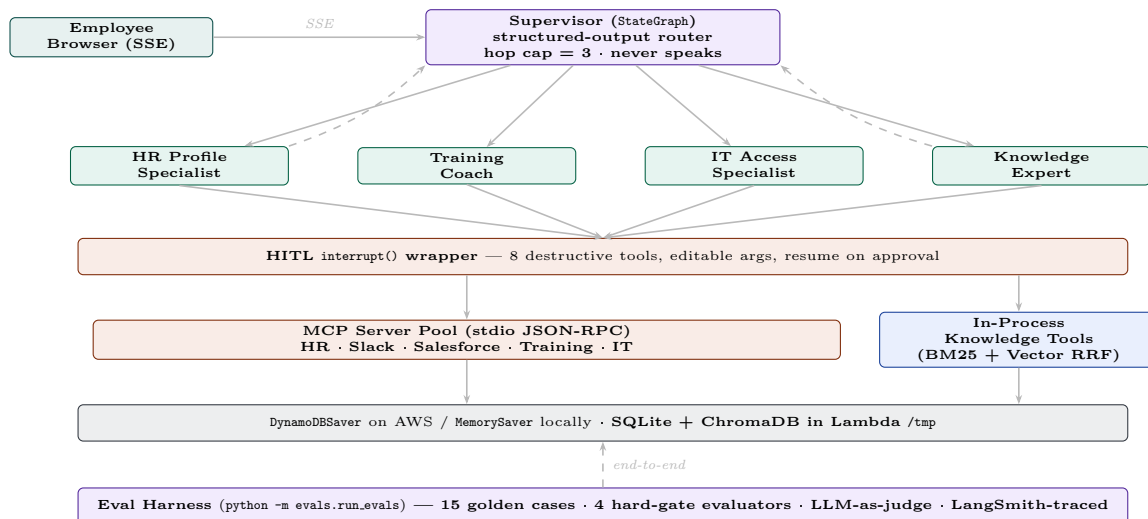
Scalability

The DynamoDB checkpointer supports Lambda process churn, but the mock SQLite systems do not provide shared horizontal state. The BM25 index rebuilds at startup — fine for a static knowledge corpus; a dynamic knowledge base needs incremental indexing and a managed vector store. Real SaaS integrations should replace the embedded mock database.

Security & Governance

- `employee_id` selects mock data; it is not identity or authorization.
- Sensitive fields (salary, SSN) should be excluded via field-level access control before reaching LLM context.
- CloudWatch receives structured logs; production audit events should be shipped to a SIEM with PII controls.
- The API and `/api/admin/reset-db` require authentication, RBAC, and rate limiting before real data is used.
- Reserved concurrency is disabled for the current account quota, so it is not a spend cap. Add Budgets/alarms and throttling; Bedrock, Lambda, the warmer, logs, ECR, DynamoDB, S3, and CloudFront can incur charges.

Supervisor Graph · HITL · Evals



The supervisor is a pure router; each specialist is a `create_react_agent` subgraph with a *scoped* toolbox, so tool choice stays tight by construction. Eight destructive tools pass through an `interrupt()` wrapper that surfaces an editable approval card before a write; DynamoDB checkpoints the paused graph on AWS. The public demo runs behind CloudFront with a response-streaming Lambda backend, Bedrock models, and Terraform/GitHub Actions delivery. A 15-case golden dataset gates CI on routing, tool-trajectory, tool-choice, and response-contains failures.

Stack: Python 3.13 · FastAPI · LangGraph (supervisor + HITL) · LangChain · ChromaDB · FastMCP · Next.js 16 · AWS Lambda · Bedrock · DynamoDB